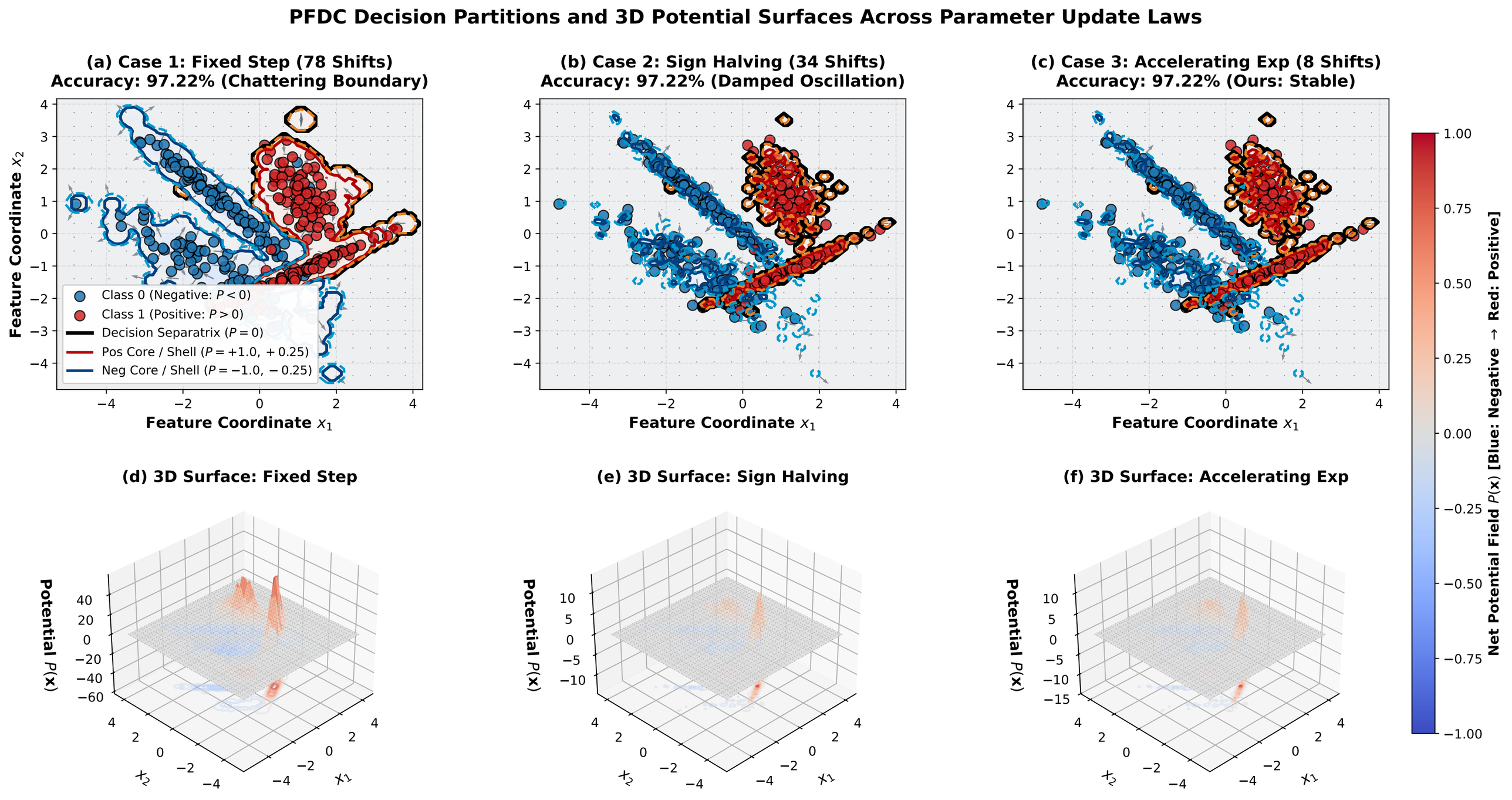

PFDC Decision Partitions and 3D Potential Surfaces Across Parameter Update Laws
(a) Case 1: Fixed Step (78 Shifts)
Accuracy: 97.22% (Chattering Boundary)
(b) Case 2: Sign Halving (34 Shifts)
Accuracy: 97.22% (Damped Oscillation)
(c) Case 3: Accelerating Exp (8 Shifts)
Accuracy: 97.22% (Ours: Stable)
Feature Coordinate x_1
Feature Coordinate x_2
Class 0 (Negative: $P < 0$)
Class 1 (Positive: $P > 0$)
Decision Separatrix ($P = 0$)
Pos Core / Shell ($P = +1.0, +0.25$)
Neg Core / Shell ($P = -1.0, -0.25$)
(d) 3D Surface: Fixed Step
(e) 3D Surface: Sign Halving
(f) 3D Surface: Accelerating Exp
Potential $P(\mathbf{x})$
Net Potential Field $P(\mathbf{x})$ [Blue: Negative → Red: Positive]